2021春高数下期中试题（回忆版）.

1. Multiple Choice Questions: (only one correct answer for each of the following questions.)

(1) If f is differentiable, and $z=z(x,y)$ is determined by $f(x-az, y-bz)=0$, then $a\frac{\partial z}{\partial x}+b\frac{\partial z}{\partial y}=$

(A) 1 (B) -1 (C) a (D) b.

(2) Let $a_n>0$ for all n. Which of the following statements must be true?

(A) If $\lim_{n\to\infty} na_n=0$, then the series $\sum_{n=1}^{\infty} a_n$ converges.

(B) If $\lim_{n\to\infty} na_n=l$ and $l\neq 0$, then the series $\sum_{n=1}^{\infty} a_n$ converges.

(C) If $\lim_{n\to\infty} na_n=l$ and $l\neq 0$, then the series $\sum_{n=1}^{\infty} a_n$ diverges.

(D) None of the above statements is correct.

(3) Indentify the surface of $2x^2+y^2=z^2$.

(A) Hyperboloid of two sheets. (B) Elliptical Cone.

(C) Hyperboloid of one sheets. (D) Elliptical paraboloid

(4) If $f(x,y)=\varphi(x+y)+\varphi(x-y)+\int_{x-y}^{x+y}\psi(t)\,dt$, where φ and ψ are twice differentiable functions, then

(A) $\frac{\partial^2 f}{\partial x\partial y}=\frac{\partial^2 f}{\partial x^2}$ (B) $\frac{\partial^2 f}{\partial x\partial y}=\frac{\partial^2 f}{\partial y^2}$ (C) $\frac{\partial^2 f}{\partial x^2}=-\frac{\partial^2 f}{\partial y^2}$ (D) $\frac{\partial^2 f}{\partial x^2}=\frac{\partial^2 f}{\partial y^2}$.

(5) $\lim_{(x,y)\to(0,0)}(1+xy)^{\frac{1}{x^2+y^2}}=$

(A) 0 (B) 1 (C) e (D) does not exist.

2. Fill in the blanks.

(1) If a,b,c are unit vectors and $a+b+c=0$, then $a\cdot b+b\cdot c+c\cdot a=$ _____.

(2) If the vector c is perpendicular to $a=\langle 1,2,1\rangle$ and $b=\langle -1,1,1\rangle$ and $c\langle i+2j+k\rangle=16$, then $c=$ _____.

(3) If $\sum_{n=2}^{\infty}\left(\tan\frac{1}{n}-k\ln\left(1-\frac{1}{n}\right)\right)$ converges, then $k=$ _____.

(4) The maximum curvature k of function $y(x)=\sin x$ is _____.

(5) If $(z+y)^x=xy$, then $\frac{\partial z}{\partial x}(1,2)=$ _____.

3. Given a cardioid $r=a(1+\cos\theta)$, $a>0$ and a circle $r=a$.

(1) Find the area of the region that lies inside the cardioid and outside the circle.

(2) Find the area of the region that lies inside the cardioid and inside the circle.

4. Assume $r(t)=f(t)i+g(t)j+h(t)k$, where

$$f(t)=\int_0^t \cos(x^2)\,dx,\quad g(t)=-t\cos t,\quad h(t)=\sum_{n=1}^{\infty}\frac{t^n}{n}.$$

Calculate $r'(0)$.

5. Let $f(x,y)=\begin{cases} y\sin\frac{1}{x^2+y^2}, & (x,y)\neq(0,0)\\ 0, & (x,y)=(0,0)\end{cases}$

(1) Is $f(x,y)$ is continuous at $(0,0)$?

(2) Find $f_x(0,0)$ and $f_y(0,0)$, if they exist.

6. Find the limit, if it exists, or show that the limit does not exist.

(1) $\lim\limits_{(x,y)\to(0,0)} \dfrac{xy}{\sqrt{x^2+y^2}}$

(2) $\lim\limits_{(x,y)\to(0,0)} \dfrac{xy^3+2x^2y^4}{x^2+y^6}$

7. For the power series $f(x)=\sum\limits_{n=1}^{\infty} \dfrac{n+2}{n(n+1)}x^n$,

(1) For what values of x does the power series converge?

(2) Find the sum of the series within the interval of convergence.

8. Determine if the series, $\sum\limits_{n=1}^{\infty} \dfrac{(-1)^{n+1}}{n^p(\ln n)^2}$ $(p>0)$, converges absolutely, or converges conditionally, or diverge. Give reasons for your answer.

9. Find $\lim\limits_{n\to\infty}\left((n^2-n)e^{\frac{1}{n}}-\sqrt{n^4+1}\right)$.